

修士論文概要書

Master's Thesis Summary

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研究指導名 Research guidance	暗号プロトコル 研究	学籍番号 Student ID number	5124FG15		
研究題目 Title	When Does a Precompile Pay? A Cost Criterion for Post-Quantum Signature Verification in a Zero-Knowledge Virtual Machine				

1 Introduction

Zero-knowledge proofs demonstrate that a computation is correct without revealing its private inputs. There are two types of zero-knowledge proof system. In a zkSNARK [1], the computation to prove is defined before the system is built. Conversely, in a zkVM [2] the computation is an input, so it is flexible. An anonymous credential lets a holder prove a statement about themselves without revealing who they are. What a holder must prove differs from one verifier to the next, so a zkVM is more suitable. Showing such a credential requires verifying a signature, and this thesis uses a post-quantum signature scheme [3]. However, proving a signature's verification inside a zkVM is expensive. One signature verification did not finish in five hours on a personal laptop. This thesis makes two contributions. First, a cost criterion for signature verification. Before building a program, it decides whether a zkVM suffices or a precompile is needed to speed up the costly operations of the major primitives. Second, we run BDEC [4], a post-quantum anonymous credential, inside a zkVM, and measure its cost and which of its security guarantees survive. To our knowledge, this is the first post-quantum anonymous credential run inside a zkVM. The zkVM we study cannot make a single proof that is both post-quantum and zero-knowledge. Achieving both in one proof is left for future work.

2 The Field-Mismatch Tax

We formalize the *field-mismatch tax*. A zkVM computes natively over small numbers, so each of the far larger numbers used by PLUM [3], the post-quantum signature scheme this thesis uses, is split across many machine words. A single multiplication must then be carried out across those parts, and its cost rises with the square of their number. Because a proof's cost tracks the total work performed, the tax is a growing penalty.

This yields the criterion. Whether a *precompile* is worth building depends on two comparisons. First, compared with the native method (without precompile), it is worth building a precompile when the work it removes exceeds the work its own circuit adds. Second, compared with an already available precompile for the same operation, a new one is worth building only when it adds less work to prove than that one does (§3).

3 The Precompile Suite

A *precompile* is a custom accelerator for a single operation that the program calls directly, so the operation runs at full speed instead of being emulated. PLUM verification has three operations heavy enough to warrant one: a hash, a large-number multiplication, and the pseudorandom check at PLUM's core. We build an accelerator for each in SP1 [5], the zkVM this thesis uses, over PLUM's 199-bit numbers, each with a written soundness argument, using the Griffin hash [6] for the first. The accelerated layer preserves PLUM's unforgeability so long as each accelerator faithfully computes its operation and the zkVM binds it to the program (§5). The criterion then separates the three. The Griffin accelerator restores feasibility (§4) yet still proves slower than a plain software SHA-3. The multiplication accelerator is predicted not to pay, because the zkVM already ships a multiplier that serves it. The pseudorandom check, the one operation that cannot be moved to smaller numbers, does pay. It runs $10.33\times$ cheaper by removing the per-step overhead of the slow version.

4 Evaluation

We compare four proving runs, which we call *cells* (Table 1). Cells 1–3 all prove PLUM at $\lambda = 80$. Cell 1 runs the Griffin hash without the accelerator, Cell 2 with it, and Cell 3 is a SHA-3 control. We use $\lambda = 80$ because it is the highest level that completes on this hardware. The full 128 level is untested and expected to run out of memory. Cell 4 (Aurora [7] proving Loquat [8]) is a different scheme over different numbers and is the only one that hides its inputs, so it serves only as a rough cross-scheme reference.

Table 1. Four cells (24 GB; cells 1–3: PLUM, $\lambda=80$, $n=5$ for cells 2–3; cell 4: Loquat in Aurora, target 80-bit, $n=3$).

Cell	Setting	Prove
1	no precompile	> 5 h
2	Griffin precompile	14.24 min
3	SHA-3 control	13.24 min
4	Aurora (Loquat)	≈ 22 min

With the accelerator, verification finishes for the first time in the study (using at most ≈ 13.7 GiB). Cell 1 stayed within memory but was stopped, unfinished, after five hours. On cost, the control is the sharpest result. Cell 3 is a diagnostic only, not a secure form of PLUM, because PLUM's security analysis does

not cover SHA-3. Even with the accelerator, PLUM verification still proves $1.08\times$ slower, and runs about $1.11\times$ more steps, than the plain software SHA-3, so a standard hash beats the accelerated one. The criterion predicts this. The accelerator removes the emulation cost, but its own circuit must still be proved. A separate control, in which the same multiplication carried out in one part rather than seven costs $164\times$ less, confirms the penalty comes from splitting numbers across parts. Both halves of the credential, issuing it and showing it, also run and prove on the zkVM (≈ 69 min and ≈ 103 – 137 min), so the study covers the whole credential, not one signature alone.

5 Discussion

Proving is now feasible but not yet deployable, and the gap is a *dual obstruction*. The proof that is both feasible and post-quantum, the plain *receipt* the zkVM emits, does not hide the holder’s data (the first obstruction). The wrap SP1 adds to hide it does fit the 24 GB budget, but it relies on elliptic-curve cryptography, which a quantum computer breaks, so it is not post-quantum (the second obstruction). A future quantum adversary could later unmask the holder of a stored proof. No proof SP1’s tools produce both hides its data and resists a quantum computer. The gap lies in the wraps SP1 ships, not in zkVMs as such. For the credential BDEC [4] built on PLUM, we prove a sharper limit. This route cannot give *everlasting* anonymity (safety even against an attacker who stores a proof today and breaks it later with unlimited power), unless the proof first conceals its data with an extra step the toolkit does not provide [9]. The positive security results are conditional, resting on five named open assumptions together with standard ones inherited from the components.

6 Conclusion

We formalized the field-mismatch tax and a criterion for when a custom precompile is worth building. We ran the first post-quantum anonymous credential inside a zkVM, within a personal machine’s 24 GB budget. And we found, with SP1’s shipped tools, that no single proof both resists a quantum computer and hides its data. Closing this gap is left to future work. One route is to make the post-quantum proof hide its own data with a hash-based method [10, 11], rather than the elliptic-curve wrap that brings the quantum weakness back.

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